

1. What is the output of a two-input AND gate when both inputs are 1?

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|---------------|--------------------|
| (A) 1 | (B) 0 |
| (C) Undefined | (D) High Impedance |

2. According to the Idempotent Law, what does $A + A$ simplify to?

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|-------|----------|
| (A) A | (B) $2A$ |
| (C) 1 | (D) 0 |

3. According to the Identity Law, what is the result of $A \cdot 1$?

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|-------|----------|
| (A) A | (B) 1 |
| (C) 0 | (D) A' |

4. What is the Boolean expression for a NOT gate applied to input A?

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|-----------------|-------------|
| (A) A' | (B) $A + 1$ |
| (C) $A \cdot 0$ | (D) $A + A$ |

5. According to De Morgan's First Theorem, what is the complement of $(A + B)$?

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|-------------------|---------------|
| (A) $A' \cdot B'$ | (B) $A' + B'$ |
| (C) $A \cdot B$ | (D) $A' + B$ |

6. According to the Complement Law, what is the result of $A + A'$?

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| (A) 1 | (B) 0 |
| (C) A | (D) A' |

7. What is the output of an OR gate if both inputs are 0?

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|------------|--------------------|
| (A) 0 | (B) 1 |
| (C) Toggle | (D) High Impedance |

8. According to the Boundedness (Null) Law, what is $A + 1$ equal to?

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|-------|----------|
| (A) 1 | (B) A |
| (C) 0 | (D) A' |

9. According to the Involution Law, what is double negation $(A')'$ equal to?

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|-------|----------|
| (A) A | (B) A' |
| (C) 0 | (D) 1 |

10. According to the Complement Law, what is the value of $A \cdot A'$?

- (A) 0
- (B) 1
- (C) A
- (D) A'

11. According to De Morgan's Second Theorem, what is the complement of $(A \cdot B)$?

- (A) $A' + B'$
- (B) $A' \cdot B'$
- (C) $A + B$
- (D) $A' \cdot B$

12. Which logic gate produces an output of 1 only when both inputs are different?

- (A) XOR
- (B) XNOR
- (C) AND
- (D) OR

13. What is the dual of the Boolean expression $A + 0 = A$?

- (A) $A \cdot 1 = A$
- (B) $A' + 1 = A'$
- (C) $A \cdot 0 = 0$
- (D) $A + 1 = 1$

14. Simplify the Boolean expression: $A + A \cdot B$.

- (A) A
- (B) B
- (C) $A + B$
- (D) $A \cdot B$

15. How many rows are in the truth table for a 3-variable Boolean function?

- (A) 8
- (B) 6
- (C) 4
- (D) 16

16. Which gate is considered a "universal gate" because any Boolean function can be implemented using only this gate type?

- (A) NAND
- (B) AND
- (C) OR
- (D) XOR

17. According to the Distributive Law, what does $A \cdot (B + C)$ expand to?

- (A) $A \cdot B + A \cdot C$
- (B) $(A+B) \cdot (A+C)$
- (C) $A + B + C$
- (D) $A \cdot B \cdot C$

18. What is the output of a NOR gate when both inputs are 0?

- (A) 1
- (B) 0
- (C) Toggle
- (D) High Impedance

19. According to the Idempotent Law, what is $A \cdot A$ equal to?

- (A) A (B) A^2
(C) 1 (D) 0

20. According to the Boundedness (Null) Law, what is $A \cdot 0$ equal to?

- (A) 0 (B) A
(C) 1 (D) A'

21. Simplify the expression: $A + A' \cdot B$.

- (A) $A + B$ (B) $A \cdot B$
(C) A (D) B

22. What is the complement of the constant 0 in Boolean algebra?

- (A) 1 (B) 0
(C) Undefined (D) -1

23. Which gate produces an output that is the inverse of an XOR gate?

- (A) XNOR (B) NAND
(C) NOR (D) AND

24. Which basic law states that $A + B = B + A$?

- (A) Commutative Law (B) Associative Law
(C) Distributive Law (D) Absorption Law

25. Simplify the Boolean expression: $(A + B) \cdot (A + B')$.

- (A) A (B) B
(C) $A + B$ (D) 0

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(A)	6	(A)	11	(A)	16	(A)	21	(A)
2	(A)	7	(A)	12	(A)	17	(A)	22	(A)
3	(A)	8	(A)	13	(A)	18	(A)	23	(A)
4	(A)	9	(A)	14	(A)	19	(A)	24	(A)
5	(A)	10	(A)	15	(A)	20	(A)	25	(A)